

Mitigating the Impact of Outlier Channels for Language Model Quantization with Activation Regularization

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Abstract

We consider the problem of accurate quantization for language models, where both the weights and activations are quantized to 4 bits per parameter with uniform quantization, the lowest bitwidth format natively supported by existing GPU hardware. In this context, the key challenge is *activation quantization*: it is known that language models contain outlier channels whose values on average are orders of magnitude higher than other channels, which prevents accurate low-bitwidth quantization with known techniques. We systematically study this phenomena and find that these outlier channels emerge early in training, and that they occur more frequently in layers with residual streams. We then propose a simple strategy which regularizes a layer’s *inputs* via quantization-aware training (QAT) and its *outputs* via activation kurtosis regularization. We show that regularizing both the inputs and outputs is crucial for preventing a model’s “migrating” the difficulty in input quantization to the weights, which makes post-training quantization (PTQ) of weights more difficult. When combined with weight PTQ, we show that our approach can obtain a W4A4 model with integer quantization that performs competitively to the standard-precision W16A16 baseline.¹

1 Introduction

Large language models (LLM) have been shown to contain *outlier channels*, i.e., feature dimensions whose values are orders of magnitude higher than the others. These outlier channels are known to be highly correlated with strong model performance (Kovaleva et al., 2021; Puccetti et al., 2022), but pose significant challenges from a model compression perspective, for instance via post-training quantization (PTQ) (Dettmers et al., 2022; Xiao et al., 2023; Wei et al., 2022). Concretely, to enable the use of low-bitwidth integer matrix multiplications—which can lead to significant speed-ups—both the activations and the weights need to be quantized. However the presence of high outlier values in the model activations results in high quantization errors, and thus overall poor PTQ accuracy (see, e.g., Xiao et al. (2023)).

To mitigate the effect of outlier channels for activation quantization at the per-tensor level, existing works have explored various approaches, including keeping some of the computations in higher precision (Dettmers et al., 2022; Ashkboos et al., 2023; Zhao et al., 2023), or “migrating” the difficulty of quantizing outlier channels to other parts of the model (Xiao et al., 2023; Wei et al., 2023; Liu et al., 2023). While the above strategies have been effective for achieving INT8 activation quantization, INT4 quantization with PTQ methods remains an open challenge, with current methods still facing nontrivial degradations in perplexity (Wu et al., 2023; Shao et al., 2023; Yuan et al., 2023).

In this work, we perform an empirical study of outlier channel phenomena from a pre-training perspective. We find that dimensions with outlier channels emerge relatively early in training (see fig. 1(a), top), suggesting that their mitigation requires early intervention. These outlier channels are particularly prevalent in the output projection layer of the first

¹Code is available at <https://github.com/aninrusimha/qat-pretrain>